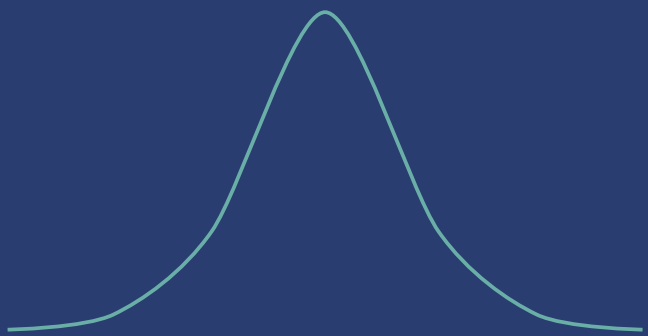
AN INTRODUCTION TO PROBABILITY AND STATISTICS

Chance, Data, and Decisions

A visual, inquiry-driven course with exercises





One-semester course packet

events → random variables → samples → inference

How to use this course

Big idea

Probability starts with a model of what might happen; statistics works backward from what did happen. The course repeatedly follows one cycle:

$$\text{model} \longrightarrow \text{simulate} \longrightarrow \text{observe} \longrightarrow \text{infer} \longrightarrow \text{check}.$$

The point is not merely to calculate a probability or a confidence interval, but to understand which assumptions made that calculation meaningful.

Audience and prerequisites

This packet is designed for a first undergraduate course or a mathematically ambitious secondary course. Students should be comfortable with algebra, functions, summation notation, and basic differentiation and integration. Multivariable calculus is not required. Proofs are included, but the emphasis is on modeling, interpretation, and the disciplined use of computation.

Learning goals

By the end, students should be able to:

1. translate verbal questions into sets, events, and probability models;
2. use counting, conditioning, independence, and Bayes' rule correctly;
3. distinguish permutations from combinations, prove basic binomial identities, and apply inclusion–exclusion to both counts and probabilities;
4. derive the birthday-paradox formula from an explicit probability space and distinguish any-pair matches from matches with one designated person;
5. move fluently among CDFs, probability masses, and densities;
6. compute and interpret expectation, variance, covariance, and correlation;
7. recognize standard distributions and justify an appropriate model;
8. obtain marginal and conditional distributions from a joint model;

9. explain the law of large numbers and central limit theorem through both mathematics and simulation;
10. evaluate estimators by bias, standard error, and mean squared error;
11. construct and interpret basic confidence intervals and hypothesis tests;
12. perform a simple Bayesian update and compare credible and confidence intervals;
13. distinguish exchangeability from independence and explain how a shared random parameter produces dependence and a random limiting frequency.

A 14-week route

The design assumes two 75-minute meetings per week. "Lab" means that students make a prediction, run an experiment or simulation, and explain a discrepancy.

Week	Core mathematics	Suggested lab or assessment
1	Sets, events, Venn diagrams, counting	Inclusion–exclusion gallery
2	Probability spaces and axioms	Build and critique sample spaces
3	Conditional probability and trees	Medical-test simulation
4	Independence, total probability, Bayes	Monty Hall and base-rate lab
5	Random variables, CDFs, PMFs, densities	One distribution in three representations
6	Expectation, variance, moments	Indicator-variable studio
7	Bernoulli, binomial, geometric, Poisson	Rare-event approximation lab
8	Uniform, exponential, normal, beta	Memorylessness and normal areas
9	Joint, marginal, conditional laws	Heat maps and independence
10	Covariance, correlation, conditional expectation	Correlation counterexamples
11	Laws of large numbers; Monte Carlo	Running-average experiment
12	Central limit theorem; sampling distributions	Visual CLT; Problem Set 5
13	Estimation, confidence intervals, tests	Coverage and power simulation
14	Bayesian inference and synthesis	Prior-to-posterior capstone

Assessment design

Optional synthesis. Chapter 10 follows Bayesian inference as an exchangeability capstone. Its coin and urn examples can anchor a week-14 synthesis session; the optional proof is extension reading, not an added prerequisite for the first weeks. A short preview in Chapter 3 plants the distinction between symmetry and independence.

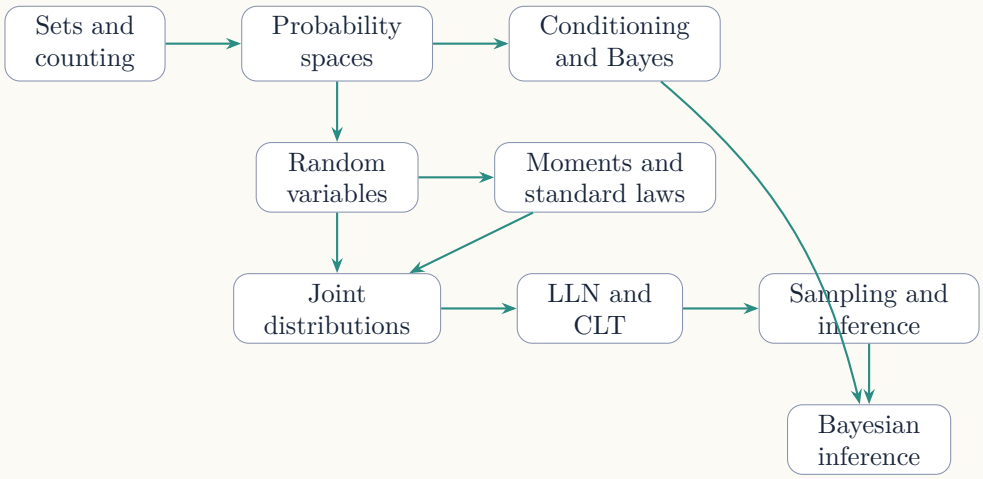
Early-course companion. Chapters 1 and 2 now provide a fuller development of permutations, combinations, binomial coefficients, both forms of inclusion–exclusion, finite probability spaces, and the birthday paradox. The companion Problem Set 2 revisits the probability spaces introduced in week two and connects them to this counting material. Its worked solutions are supplied separately for the instructor.

One workable balance is 35% problem sets, 20% short quizzes, 20% computational labs, and 25% a modeling capstone. Exercises use three labels:

- **Fluency**: calculations and translations that make later reasoning automatic.

- **Reasoning**: derivations, short proofs, and careful interpretation.
- **Exploration**: simulation, visualization, model criticism, or open-ended comparison.

Dependency map



Checkpoint

A good opening prompt. A family has two children and at least one is a girl. What is the probability that both are girls? Ask students to defend not just an answer, but a sample space. Then change how the information was obtained and ask whether the answer changes.

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Chapter 1

The grammar of uncertainty

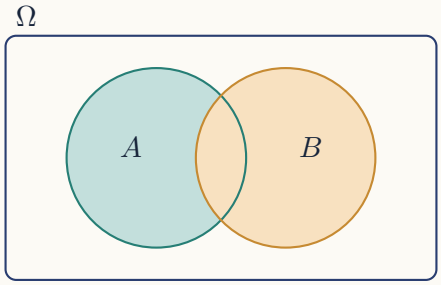
Sets, permutations, combinations, and inclusion–exclusion

Guiding question 1.1. Before assigning probabilities, how do we describe precisely what could happen?

A *sample space* is built from possible outcomes; an *event* is a collection of outcomes. Set notation is therefore the grammar of probability.

1.1 Set operations

Let Ω be a universe of outcomes and $A, B \subseteq \Omega$. The union $A \cup B$ means "A or B (or both)," the intersection $A \cap B$ means "both," and the complement A^c means "not A." The difference $A \setminus B$ contains outcomes in A but not B.



Overlap represents $A \cap B$

De Morgan's laws translate negated statements:

$$(A \cup B)^c = A^c \cap B^c, \qquad (A \cap B)^c = A^c \cup B^c.$$

Common trap

In everyday speech, "or" is sometimes exclusive. In probability, $A \cup B$ is inclusive unless explicitly stated otherwise.

1.2 Products and sequences

The Cartesian product $A \times B$ contains ordered pairs (a, b). Repeated trials naturally live in product spaces. Three coin flips have sample space $\{H, T\}^3$, with eight ordered strings. Order matters: HTT and THT are different outcomes.

1.3 The multiplication principle

If a first choice has m possibilities and, after *each* first choice, a second choice has n possibilities, there are mn ordered outcomes. More generally, if the successive numbers of choices are $n_1, \ldots, n_k$, the number of outcomes is $n_1 \cdots n_k$. The available choices may depend on earlier choices; their *number* at each stage must be the stated one. If different first choices leave different numbers of continuations, count each branch and add its contributions instead.

Example 1.2 (Codes and officers)**.** A four-digit code may begin with zero. If digits can repeat, there are 10^4 codes. If digits must be distinct, the count is $10 \cdot 9 \cdot 8 \cdot 7 = 5040$. Assigning president, secretary, and treasurer from ten students similarly gives $10 \cdot 9 \cdot 8 = 720$ outcomes when no student holds two offices.

1.4 Permutations and combinations

There are $n! = n(n-1)\cdots 1$ orders of n distinct objects, with $0! = 1$. For $0 \le k \le n$, an ordered selection of k distinct objects from n has

$$(n)_k = n(n-1)\cdots(n-k+1) = \frac{n!}{(n-k)!}$$

possibilities. This is sometimes written $P(n, k)$. An unordered selection of k objects is a *combination*. Every such selection has exactly $k!$ orderings, so

$$\binom{n}{k} = \frac{(n)_k}{k!} = \frac{n!}{k!(n-k)!}.$$

In particular, $\binom{n}{0} = \binom{n}{n} = 1$: there is one empty selection and one selection containing everything. We use $\binom{n}{k} = 0$ when $k < 0$ or $k > n$ and n is a nonnegative integer.

Experiment	What distinguishes outcomes?	Count
k draws with replacement	Order matters; repetition allowed	n^k
k draws without replacement	Order matters; objects distinct	$(n)_k$
A k-element subset	Order ignored; no repetition	$\binom{n}{k}$

Example 1.3 (A committee is not a list of officers)**.** From ten students there are $\binom{10}{3} = 120$ three-person committees, but 720 assignments to three distinct offices. A committee with one designated chair has $3\binom{10}{3} = 360$ possibilities: the two other members have no distinct roles. The question about order must be answered before choosing a formula.

Proposition 1.4 (Arrangements with repeated objects)**.** *If n objects consist of r types with multiplicities $n_1, \ldots, n_r$, where $n_1 + \cdots + n_r = n$, the number of distinct arrangements is $n!/(n_1! \cdots n_r!)$.*

Proof. Temporarily distinguish objects of the same type with labels. There are $n!$ arrangements of the labeled objects. Each arrangement after erasing the labels comes from exactly $n_1! \cdots n_r!$ labeled arrangements. Dividing by that constant multiplicity gives the result. □

For example, `BANANA` has $6!/(3!2!) = 60$ distinct arrangements.

1.5 Binomial coefficients count in two directions

Proposition 1.5 (Symmetry and Pascal's identity)**.** *For* $0 \le k \le n$,

$$\binom{n}{k} = \binom{n}{n-k},$$

and, for $n \ge 1$,

$$\binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}.$$

Proof. A k-element subset determines its $(n-k)$-element complement, proving symmetry. For Pascal's identity, single out one object. A k-element subset either contains it, leaving $k-1$ choices from the other $n-1$ objects, or omits it, leaving k choices from those objects. These two cases are disjoint and exhaustive. □

Theorem 1.6 (Binomial theorem)**.** *For a nonnegative integer* n,

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k.$$

Proof. Expand the product of n copies of $(x+y)$. To produce $x^{n-k}y^k$, choose the k factors that supply y; all the others supply x. There are $\binom{n}{k}$ choices. □

Taking $x = y = 1$ gives $\sum_{k=0}^{n} \binom{n}{k} = 2^n$. This also counts all subsets in two ways: by their sizes, or by making an include/exclude choice for each object. Taking $x = 1, y = -1$ gives, for $n \ge 1$, $\sum_{k=0}^{n} (-1)^k \binom{n}{k} = 0$. This alternating sum is the engine behind the general inclusion–exclusion formula.

1.6 Inclusion–exclusion for finite sets

Proposition 1.7 (Two and three sets)**.** *For finite sets* A, B, C,

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B|, \\ |A \cup B \cup C| &= |A| + |B| + |C| \\ &\quad - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|. \end{aligned}$$

For two sets, subtraction repairs the double count of the overlap. With three sets, an element in all three is added three times and subtracted three times, so it must be added back once. An element in exactly two sets contributes $2 - 1 = 1$; an element in exactly one contributes 1. An element outside the union contributes zero. This proves the formulas.

Common trap

$A \cap B$ includes the triple intersection $A \cap B \cap C$. It does not mean "in A and B but outside C." Confusing these two regions produces incorrect inclusion–exclusion calculations.

Example 1.8 (Three activities)**.** Among 100 students, 39 attend activity A, 29 attend B, and 25 attend C. The pairwise intersection counts are 13, 11, and 9, respectively, and 5 students attend all three. Thus the union has $39 + 29 + 25 - 13 - 11 - 9 + 5 = 65$ students; 35 attend none. The exactly-two count is $(13 - 5) + (11 - 5) + (9 - 5) = 18$. The exactly-one count is $65 - 18 - 5 = 42$.

Theorem 1.9 (Finite inclusion–exclusion)**.** *For finite sets $A_1, \ldots, A_r$,*

$$\left|\bigcup_{i=1}^{r} A_i\right| = \sum_{\varnothing \neq J \subseteq \{1,\ldots,r\}} (-1)^{|J|+1} \left|\bigcap_{j \in J} A_j\right|.$$

Equivalently: add the single-set sizes, subtract the pairwise intersection sizes, add the triple-intersection sizes, and continue with alternating signs through the intersection of all r sets.

Proof. An element belonging to exactly $t \geq 1$ sets occurs in $\binom{t}{k}$ of the k-fold intersections. Its total contribution is

$$\sum_{k=1}^{t} (-1)^{k+1} \binom{t}{k} = 1 - (1-1)^t = 1.$$

An element in no set contributes zero. Thus each element of the union is counted exactly once. □

Example 1.10 (Using all three symbols)**.** There are 3^4 strings of length four over $\{A, B, C\}$. To count those using every symbol, subtract strings missing at least one symbol. Each missing-symbol event has 2^4 strings; each pair of missing-symbol events has one string; no length-four string misses all three symbols. Therefore the required count is

$$3^4 - 3 \cdot 2^4 + 3 \cdot 1^4 = 36.$$

These are counts; turning them into probabilities requires a specified probability model.

Example 1.11. A five-card poker hand is an unordered choice from 52 cards, so there are $\binom{52}{5}$ hands. Exactly two aces means choosing two of four aces and three of the remaining 48 cards:

$$\binom{4}{2}\binom{48}{3}.$$

Explore

Use the website's counting lab to compare committees, ordered selections, and repeated draws. Then use the inclusion–exclusion lab to vary the eight disjoint regions of three sets, including the region outside the union. Recover the marginal and intersection totals from those regions. With all seven marginal/intersection totals specified, the seven interior region counts are determined; knowing the universe size also determines the outside region. If totals are omitted, the remaining data need not determine a unique diagram.

Exercises

1. Write $(A \cap B^c) \cup (A^c \cap B)$ in words. [Fluency]

2. Verify both De Morgan laws using membership tables. [Reasoning]

3. How many eight-character strings contain exactly three letters and five digits? [Fluency]

4. Count five-card hands containing at least one ace. [Reasoning]

5. Prove $\binom{n}{k} = \binom{n}{n-k}$ by a counting argument. [Reasoning]

6. Among integers from 1 to 1,000, how many are divisible by 2 or 3 or 5? [Reasoning]

7. Design two different sampling procedures that produce the same set of possible outcomes but not the same probabilities. [Exploration]

8. From nine students, count a four-person committee, a list of four distinct officers, and a four-person committee with one chair. [Fluency]

9. Count the distinct arrangements of the letters in `LEVEL`. Explain the divisor in your formula. [Fluency]

10. Prove Pascal's identity by deciding whether a distinguished object is selected. [Reasoning]

11. Find the coefficient of x^2 in $(1 + 3x)^5$, and explain its binomial coefficient. [Fluency]

12. Count length-five strings over $\{A, B, C\}$ that use all three symbols. [Reasoning]

Chapter 2

A measure of possibility

Probability spaces, inclusion–exclusion, and the birthday paradox

Guiding question 2.1. What rules must any coherent numerical description of uncertainty obey?

2.1 The axioms

Definition 2.2. A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$: a sample space Ω, a collection $\mathcal{F}$ of events containing Ω and closed under complements and countable unions, and a function $\mathbb{P} : \mathcal{F} \to [0,1]$ satisfying

1. $\mathbb{P}(\Omega) = 1$;

2. if $A_1, A_2, \ldots$ are pairwise disjoint, then

$$\mathbb{P}\left(\bigcup_i A_i\right) = \sum_i \mathbb{P}(A_i).$$

The collection $\mathcal{F}$ is called a *sigma-algebra*. For a finite sample space we usually take $\mathcal{F} = 2^\Omega$, the collection of all subsets of Ω. The more careful notation becomes important for continuous spaces.

2.2 Building a finite probability space

Choose nonnegative numbers w_ω for the individual outcomes such that $\sum_{\omega\in\Omega} w_\omega = 1$, and define

$$\mathbb{P}(A) = \sum_{\omega \in A} w_\omega.$$

The total mass is one; for disjoint events, the sums use disjoint outcomes, which gives additivity. Since Ω is finite, only finitely many events in a disjoint family can be nonempty, so countable additivity also follows. This construction gives every probability measure on a finite space with all subsets measurable. The uniform model is the special choice $w_\omega = 1/|\Omega|$ for each outcome.

Example 2.3 (Unequal weights)**.** Let $\Omega = \{a, b, c, d\}$ have masses $1/10, 2/10, 3/10, 4/10$. For $A = \{a, b\}$ and $B = \{b, d\}$,

$$\mathbb{P}(A) = \frac{3}{10}, \quad \mathbb{P}(B) = \frac{6}{10}, \quad \mathbb{P}(A \cap B) = \frac{2}{10}, \quad \mathbb{P}(A \cup B) = \frac{7}{10}.$$

Although A contains half the outcomes, it does not have probability $1/2$. Counting outcomes and adding their probability masses are different operations unless all outcomes have equal mass.

Proposition 2.4. *For events A, B,*

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \qquad \mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

If $A \subseteq B$, then $\mathbb{P}(A) \leq \mathbb{P}(B)$.

Each statement follows by splitting a set into disjoint pieces. This is the basic proof move in probability.

The two-event formula. Write $A \cup B = A \sqcup (B \setminus A)$, where $\sqcup$ denotes disjoint union. Also $B = (A \cap B) \sqcup (B \setminus A)$. Additivity gives $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A \cap B)$; substitute the second identity into the first. □

2.3 Inclusion–exclusion for probabilities

For any three events, the same overlap correction gives

$$\begin{aligned}\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) \\ - \mathbb{P}(A \cap B) - \mathbb{P}(A \cap C) - \mathbb{P}(B \cap C) + \mathbb{P}(A \cap B \cap C).\end{aligned}$$

No independence or equal-likelihood assumption is needed. One proof applies the two-event formula to $(A \cup B) \cup C$, then to $(A \cap C) \cup (B \cap C) = (A \cup B) \cap C$.

Theorem 2.5 (Finite probabilistic inclusion–exclusion)**.** *For events $A_1, \ldots, A_r$ in any probability space,*

$$\mathbb{P}\left(\bigcup_{i=1}^{r} A_i\right) = \sum_{\varnothing \neq J \subseteq \{1,\ldots,r\}} (-1)^{|J|+1} \mathbb{P}\left(\bigcap_{j \in J} A_j\right).$$

Proof. Partition the sample space by the exact list of events that occur. There are at most 2^r such disjoint measurable regions. A region in exactly $t \geq 1$ events contributes its probability mass with coefficient $\sum_{k=1}^{t} (-1)^{k+1} \binom{t}{k} = 1$ on the right. The region in no event contributes zero. Finite additivity therefore gives the probability of the union, even when the original sample space is infinite. □

Example 2.6 (Counts become probabilities under a uniform model)**.** Choose a student uniformly from the 100 students in the activity example in Chapter 1. Dividing all its counts by 100 gives

$$\mathbb{P}(A \cup B \cup C) = .39 + .29 + .25 - .13 - .11 - .09 + .05 = .65.$$

The eight disjoint regions have probabilities $.20, .12, .10, .08, .06, .04, .05, .35$ (the three single-only regions, three pair-only regions, triple region, and outside region). They are nonnegative and sum to one, so they also specify a valid eight-outcome probability space with *unequal* outcome weights.

Common trap

Inclusion–exclusion always uses the *actual* intersection probabilities. Replacing $\mathbb{P}(A \cap B)$ with $\mathbb{P}(A)\mathbb{P}(B)$ requires an additional independence assumption. The formula itself makes no such assumption. Likewise, counts divided by a total give probabilities only for a justified uniform model.

2.4 Equally likely is an assumption

If a finite sample space has N equally likely outcomes and A contains m, then $\mathbb{P}(A) = m/N$. But equality of probabilities is part of the model, not a consequence of counting.

Example 2.7. For two independent fair dice, the ordered-pair space has 36 equally likely outcomes. The possible sums $2, 3, \ldots, 12$ are not equally likely: sum 7 has six representations, while sum 2 has one.

Recording a coarser outcome carries its probabilities along: for a recorded value s, add the masses of all original outcomes producing s. Replacing the resulting weights by equal weights changes the experiment.

2.5 The birthday paradox: count the complement

Guiding question 2.8. How large must a class be before a shared birthday is more likely than not?

State the model first. There are n labeled people and $d \geq 1$ possible days. Their birthdays are independent and uniform on those days. Thus

$$\Omega = \{1, \ldots, d\}^n, \qquad \mathcal{F} = 2^\Omega, \qquad \mathbb{P}(\{\omega\}) = d^{-n}.$$

For the usual classroom model set $d = 365$: ignore leap days and seasonal variation. Independence and equal daily probabilities are modeling assumptions, not claims about every real class.

Proposition 2.9 (Exact collision probability)**.** *For $0 \leq n \leq d$, the probability $q_d(n)$ of at least one shared birthday is*

$$q_d(n) = 1 - \frac{d(d-1)\cdots(d-n+1)}{d^n} = 1 - \prod_{j=0}^{n-1} \left(1 - \frac{j}{d}\right).$$

The empty product is one, so $q_d(0) = q_d(1) = 0$. For $n > d$, $q_d(n) = 1$ by the pigeonhole principle.

Proof. Count the complement: all birthdays are distinct. There are d choices for the first birthday, $d-1$ for the second, and so on, yielding $(d)_n$ ordered birthday lists. All d^n lists are equally likely, so divide and subtract from one. Counting *at least one* matching pair directly is harder because different matching-pair events overlap. □

Class size n	10	20	22	23	30
$q_{365}(n)$	.11695	.41144	.47570	.50730	.70632

Adding a person cannot destroy an existing match, so the probability is nondecreasing. Since $q_{365}(22) < 1/2 < q_{365}(23)$, the first class size above 50% is 23.

Any pair versus one particular person

In a class of $n \geq 1$ people, the probability that someone else shares one *designated* person's birthday is

$$1 - \left(1 - \frac{1}{d}\right)^{n-1}.$$

For $d = 365, n = 23$, this is about .05857, much smaller than .50730. The designated-person question compares that person with 22 others; the any-pair question considers $\binom{23}{2} = 253$ possible pairs.

2.6 Birthday matches meet inclusion–exclusion

For three people, let E_{12}, E_{13}, E_{23} be the events that the indicated people have the same birthday. Each event has probability $1/d$. Any two of these events force all three birthdays to agree, so each pairwise intersection has probability $1/d^2$. The triple intersection is the *same event*, with probability $1/d^2$, not $1/d^3$. Therefore inclusion–exclusion gives

$$q_d(3) = \frac{3}{d} - \frac{3}{d^2} + \frac{1}{d^2} = \frac{3}{d} - \frac{2}{d^2},$$

agreeing with $1 - (1 - 1/d)(1 - 2/d)$. This illustrates why treating all matching-pair events as independent gives a wrong answer.

More generally there are $\binom{n}{2}$ matching-pair events, each with probability $1/d$. The union bound gives the useful check

$$q_d(n) \leq \min\left(1, \frac{\binom{n}{2}}{d}\right).$$

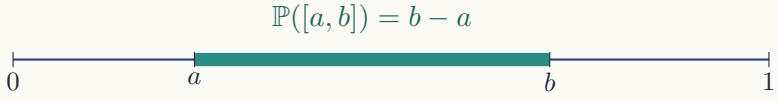
The sum $\binom{n}{2}/d$ counts overlapping pair events; it is not itself the collision probability. Later, linearity of expectation will identify it as the expected number of matching pairs.

The birthday experiment

Use the website to compare the exact product, the designated-person probability, and the union bound. Predict the 50% threshold, then simulate classes. Change the calendar from 365 days to 30 or 10 days. Separate sampling fluctuations from changes in the underlying model.

2.7 Continuity and geometry

Choose a point uniformly from $[0, 1]$. Every individual point has probability zero, yet the entire interval has probability one. Probabilities attach to regions: $\mathbb{P}([a, b]) = b - a$. Zero probability does not mean impossible; it means negligible under the model.



2.8 Simulation and the model–world gap

A probability model is a deliberate simplification. Simulation samples from the model; data come from the world. Agreement between simulation and data supports a model, but never proves it uniquely correct.

Theorem 2.10 (Union bound)**.** *For any events $A_1, \dots, A_n$,*

$$\mathbb{P}\left(\bigcup_{i=1}^{n} A_i\right) \le \sum_{i=1}^{n} \mathbb{P}(A_i).$$

The events need not be disjoint. Double counting makes the right side larger, which is harmless for an upper bound.

Exercises

1. Derive $\mathbb{P}(\varnothing) = 0$ from the axioms. [Reasoning]

2. Two independent fair dice are rolled. Find the probability that the maximum is 4. [Fluency]

3. Prove the union bound for three events. [Reasoning]

4. A point is chosen uniformly from the unit square. Find the probability that $x + y < 1$. [Fluency]

5. Give an event with probability zero that is not logically impossible. [Reasoning]

6. Estimate by simulation the probability that two random points in $[0, 1]$ are within 0.1 of each other; then derive it geometrically. [Exploration]

7. On $\Omega = \{1, 2, 3, 4\}$, set $\mathbb{P}(\{i\}) = i/10$. Compute the probability of an even outcome and compare it with the fraction of outcomes that are even. [Fluency]

8. Prove three-event probabilistic inclusion–exclusion using only the two-event formula. [Reasoning]

9. With independent uniform birthdays on 30 days, find the first class size whose collision probability exceeds $1/2$. [Fluency]

10. For four people and a five-day calendar, count the no-collision lists and calculate the collision probability. [Fluency]

11. Explain why $\binom{n}{2}/d$ can exceed one without contradicting the probability axioms. [Reasoning]

Chapter 3

Information changes the space

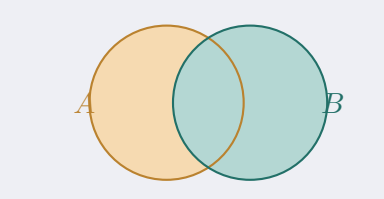
Conditional probability, independence, and Bayes' rule

Guiding question 3.1. When evidence arrives, which probabilities should change—and by how much?

Definition 3.2. If $\mathbb{P}(B) > 0$, the conditional probability of A given B is

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

Conditioning restricts attention to B and renormalizes so that B becomes the new whole.



Within B, measure the part also in A

3.1 Multiplication and trees

Rearranging the definition gives

$$\mathbb{P}(A \cap B) = \mathbb{P}(B)\mathbb{P}(A \mid B) = \mathbb{P}(A)\mathbb{P}(B \mid A).$$

Along a probability tree, multiply down branches and add across disjoint terminal paths.

3.2 Independence

Events A and B are independent when

$$\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B).$$

When probabilities are positive, this is equivalent to $\mathbb{P}(A \mid B) = \mathbb{P}(A)$. Disjointness is almost the opposite: two nontrivial disjoint events cannot be independent.

Common trap

Pairwise independence does not imply mutual independence. Three events can be independent in every pair while the third is determined by the first two.

3.3 Total probability and Bayes

If $B_1, \dots, B_k$ partition Ω, then

$$\mathbb{P}(A) = \sum_{i=1}^{k} \mathbb{P}(A \mid B_i)\mathbb{P}(B_i).$$

Bayes' rule reverses the condition:

$$\mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.$$

Example 3.3 (A diagnostic test)**.** Suppose prevalence is 1%, sensitivity is 95%, and specificity is 90%. Among 10,000 people, expect 100 cases, of which 95 test positive. Among 9,900 non-cases, 990 test positive. Thus

$$\mathbb{P}(\text{case} \mid +) = \frac{95}{95 + 990} \approx 0.0876.$$

A highly sensitive test can have a modest positive predictive value when the base rate is low.

Explore

Represent the test example with 10,000 dots rather than formulas. Change prevalence while holding sensitivity and specificity fixed. Plot the positive predictive value; locate the prevalence at which a positive result becomes more likely correct than incorrect.

3.4 A preview: symmetry is not independence

Choose once between coins with heads probabilities 1/4 and 3/4, giving each coin probability 1/2, and then repeatedly toss the chosen coin. Conditional on the choice, the tosses are independent. Without that information, if H_i means heads on toss i, then

$$\mathbb{P}(H_1) = \mathbb{P}(H_2) = \frac{1}{2}, \qquad \mathbb{P}(H_1 \cap H_2) = \frac{1}{2}\left(\frac{1}{4}\right)^2 + \frac{1}{2}\left(\frac{3}{4}\right)^2 = \frac{5}{16} \neq \frac{1}{4}.$$

The tosses are dependent, yet any reordering of a finite string has the same probability as the original string. This symmetry is called *exchangeability*. Chapter 10 returns to the example and proves de Finetti's surprising converse for infinite zero–one sequences: exchangeability gives a representation by a shared random bias and conditionally independent trials.

Exercises

1. Draw a tree for two draws without replacement from an urn with three red and two blue balls. [Fluency]

2. Prove that if A and B are independent, then A^c and B are independent. [Reasoning]

3. Construct three pairwise independent events that are not mutually independent. [Reasoning]

4. Solve the diagnostic-test example with prevalence 5%. [Fluency]

5. A randomly selected coin is fair with probability .8 and double-headed with probability .2. It lands heads three times. Find the posterior probability it is double-headed. [Reasoning]

6. Explain precisely why changing the information protocol can change the "two children" answer. [Exploration]

7. Simulate Monty Hall and explain the result using conditioning rather than frequency alone. [Exploration]

Chapter 4

Numbers attached to outcomes

Random variables, distributions, expectation, and moments

Guiding question 4.1. How can one function compress a complicated experiment into a numerical quantity of interest?

Definition 4.2. A random variable is a function $X : \Omega \to \mathbb{R}$. Its cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x).$$

Every CDF is nondecreasing, right-continuous, approaches 0 to the left and 1 to the right. For a discrete variable, jumps encode probabilities. For a continuous variable with density f, areas encode probabilities:

$$F_X(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad \mathbb{P}(a < X \leq b) = \int_a^b f(t)\,dt.$$

4.1 Expectation

For discrete X with values x_i,

$$\mathbb{E}[X] = \sum_i x_i \mathbb{P}(X = x_i).$$

For continuous X with density f,

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f(x)\,dx.$$

Expectation is a center of mass, not necessarily a possible value.

Theorem 4.3 (LOTUS)**.** *For a suitable function g,*

$$\mathbb{E}[g(X)] = \sum_x g(x)\mathbb{P}(X = x) \quad \text{or} \quad \mathbb{E}[g(X)] = \int g(x) f_X(x)\,dx.$$

Theorem 4.4 (Linearity)**.** *For constants a, b and integrable random variables X, Y,*

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y],$$

whether or not X and Y are independent.

4.2 Variance and moments

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2.$$

Standard deviation has the same units as X. The kth raw moment is $\mathbb{E}[X^k]$; the kth central moment is $\mathbb{E}[(X - \mathbb{E}X)^k]$. The moment-generating function, when finite near zero, is $M_X(t) = \mathbb{E}[e^{tX}]$ and satisfies $M_X^{(k)}(0) = \mathbb{E}[X^k]$.

Proposition 4.5. *For constants a, b,*

$$\operatorname{Var}(aX + b) = a^2 \operatorname{Var}(X).$$

4.3 Indicator variables

For an event A, let $\mathbf{1}_A = 1$ on A and 0 otherwise. Then $\mathbb{E}[\mathbf{1}_A] = \mathbb{P}(A)$. If $N = \sum_i \mathbf{1}_{A_i}$ counts occurrences, linearity gives

$$\mathbb{E}[N] = \sum_i \mathbb{P}(A_i)$$

without independence.

Theorem 4.6 (Markov and Chebyshev)**.** *If $X \geq 0$, then $\mathbb{P}(X \geq a) \leq \mathbb{E}[X]/a$. Consequently,*

$$\mathbb{P}(|X - \mathbb{E}X| \geq k\sigma) \leq \frac{1}{k^2}.$$

Exercises

1. Find the CDF of the number of heads in two fair flips. [Fluency]
2. A density is proportional to x on $[0, 2]$. Normalize it and find its mean. [Fluency]
3. Prove $\operatorname{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}X)^2$. [Reasoning]
4. Find the expected number of fixed points in a random permutation of n objects. [Reasoning]
5. Show that a fair lottery can still have a large variance. Interpret. [Exploration]
6. Use Chebyshev's inequality to bound the chance that a variable lies at least three standard deviations from its mean. Compare with the normal distribution. [Reasoning]
7. Construct two different distributions with the same mean and variance. [Exploration]

Chapter 5

A library of chance mechanisms

Standard discrete and continuous distributions

Guiding question 5.1. Which probability laws recur because the same mechanisms recur?

5.1 Discrete families

Law	Mechanism and mass function	Mean	Variance
Bernoulli(p)	one success indicator; $p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial(n,p)	successes in n independent trials; $\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$
Geometric(p)	trials through first success; $(1-p)^{k-1}p$	$1/p$	$(1-p)/p^2$
Poisson(λ)	rare-event count; $e^{-\lambda}\lambda^k/k!$	λ	λ

The geometric law is memoryless:

$$\mathbb{P}(X > s+t \mid X > s) = \mathbb{P}(X > t).$$

The Poisson law arises as a limit of $\mathrm{Bin}(n,p)$ when n is large, p is small, and $np \approx \lambda$.

5.2 Continuous families

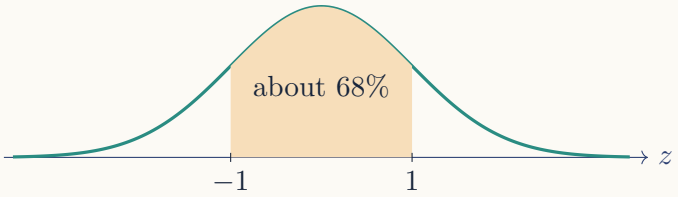
Uniform(a,b) assigns constant density $1/(b-a)$ to an interval. Exponential(λ) has density $\lambda e^{-\lambda x}$ on $x \geq 0$ and is the continuous memoryless law.

The normal density with mean μ and variance σ^2 is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right).$$

Standardization sends $X \sim \mathcal{N}(\mu,\sigma^2)$ to

$$Z = \frac{X-\mu}{\sigma} \sim \mathcal{N}(0,1).$$



The beta family on $[0,1]$ has density proportional to $x^{\alpha-1}(1-x)^{\beta-1}$. It is useful for proportions and becomes a natural prior for a Bernoulli probability.

5.3 Choose by mechanism, not by silhouette

A bell-shaped histogram does not prove normality. Ask what generated the observations, what values are possible, whether trials are independent, whether rates are stable, and what approximation error is acceptable.

Exercises

1. Compute $\mathbb{P}(X \geq 2)$ for $X \sim \text{Bin}(5, .3)$. [Fluency]
2. Prove the geometric memoryless property. [Reasoning]
3. Approximate $\text{Bin}(1000, .002)$ by a Poisson law and compare $\mathbb{P}(X = 0)$. [Exploration]
4. Find the median of an exponential distribution. [Fluency]
5. If $X \sim \mathcal{N}(10, 9)$, express $\mathbb{P}(8 < X < 13)$ using the standard normal CDF. [Fluency]
6. Derive the mean and variance of $\text{Uniform}(a, b)$. [Reasoning]
7. For each family in the chapter, invent one plausible and one implausible real-world use. [Exploration]

Chapter 6

Variables together

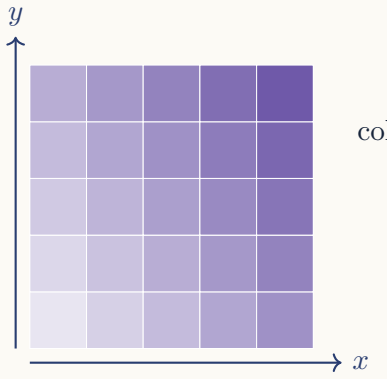
Joint, marginal, and conditional distributions

Guiding question 6.1. How do we describe dependence without confusing association, causation, and prediction?

For discrete variables, a joint mass function $p(x, y) = \mathbb{P}(X = x, Y = y)$ is a table whose entries sum to one. Marginals are obtained by summing:

$$p_X(x) = \sum_y p(x, y), \qquad p_Y(y) = \sum_x p(x, y).$$

For a continuous joint density $f(x, y)$, replace sums with integrals.



color = joint probability

6.1 Conditional laws and independence

For $p_Y(y) > 0$,

$$p_{X|Y}(x \mid y) = \frac{p(x, y)}{p_Y(y)}.$$

Variables are independent when their joint law factors:

$$p(x, y) = p_X(x)p_Y(y)$$

or $f(x, y) = f_X(x)f_Y(y)$ throughout the support.

6.2 Covariance and correlation

$$\operatorname{Cov}(X,Y) = \mathbb{E}[(X-\mathbb{E}X)(Y-\mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Correlation is the dimensionless quantity

$$\rho_{X,Y} = \frac{\operatorname{Cov}(X,Y)}{\sigma_X \sigma_Y}.$$

It lies between -1 and 1. Independence implies zero covariance when moments exist, but zero covariance does not generally imply independence.

Theorem 6.2.

$$\operatorname{Var}(X+Y) = \operatorname{Var}(X) + \operatorname{Var}(Y) + 2\operatorname{Cov}(X,Y).$$

For independent variables, the covariance term vanishes.

6.3 Conditional expectation

$\mathbb{E}[X \mid Y]$ is itself a random variable: for each observed value of Y, it gives the mean of the corresponding conditional distribution. The tower property says

$$\mathbb{E}[\mathbb{E}[X \mid Y]] = \mathbb{E}[X].$$

The conditional mean is also the best mean-square predictor of X using only Y.

Common trap

Correlation measures linear association. A perfect U-shaped relationship can have correlation zero. And neither correlation nor conditioning alone establishes causation.

Exercises

1. Given a 2×3 joint probability table, compute both marginals and one conditional distribution. [Fluency]
2. Prove that independence implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$. [Reasoning]
3. Let X be uniform on $\{-1, 0, 1\}$ and $Y = X^2$. Show $\operatorname{Cov}(X,Y) = 0$ although X, Y are dependent. [Reasoning]
4. Derive the variance-of-a-sum identity. [Reasoning]
5. If X, Y are independent Poisson variables, use generating functions to show $X+Y$ is Poisson. [Reasoning]
6. Create three scatterplots with correlations near -1, 0, and 1, then create a nonlinear plot with correlation near zero. [Exploration]
7. Explain the tower property using a population divided into groups. [Exploration]

Chapter 7

From repetition to regularity

Laws of large numbers, the central limit theorem, and sampling

Guiding question 7.1. Why do averages stabilize, and why do bell curves emerge from non-normal data?

Let $X_1, \dots, X_n$ be independent copies of a variable with mean μ and variance σ^2. Their average is

$$\bar{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i, \qquad \mathbb{E}[\bar{X}_n] = \mu, \qquad \operatorname{Var}(\bar{X}_n) = \frac{\sigma^2}{n}.$$

Theorem 7.2 (Weak law of large numbers)**.** *For every $\varepsilon > 0$,*

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \longrightarrow 0.$$

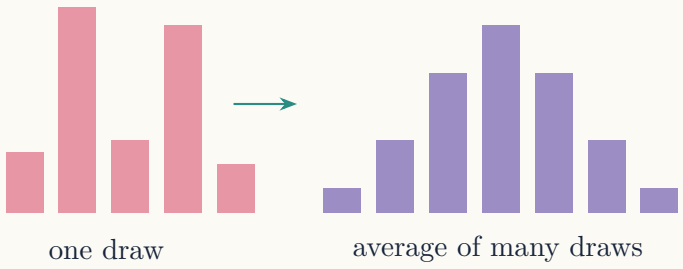
With finite variance, Chebyshev gives the proof immediately:

$$\mathbb{P}(|\bar{X}_n - \mu| > \varepsilon) \leq \frac{\sigma^2}{n\varepsilon^2}.$$

Theorem 7.3 (Central limit theorem)**.** *If $0 < \sigma^2 < \infty$, then*

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1).$$

The LLN says where the average goes; the CLT describes the scale and approximate shape of its remaining fluctuations.



7.1 Sampling distributions

A *statistic* is a function of the sample. Before data are observed it is random, and its probability law is its sampling distribution. The standard deviation of that distribution is the *standard error*.

For a Bernoulli sample, the sample proportion $\hat{p}$ has

$$\mathbb{E}[\hat{p}] = p, \qquad \mathrm{SE}(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}.$$

7.2 Monte Carlo and bootstrap

Monte Carlo approximates an expectation by simulating from a known model. The bootstrap approximates a sampling distribution by repeatedly resampling, with replacement, from observed data. The bootstrap is powerful but cannot repair a badly biased or unrepresentative original sample.

Exercises

1. Compute the standard error of the mean when $\sigma = 12$ and $n = 64$. [Fluency]
2. Prove the weak law using Chebyshev's inequality. [Reasoning]
3. Approximate $\mathbb{P}(480 \le X \le 520)$ for $X \sim \mathrm{Bin}(1000, .5)$ using the CLT and a continuity correction. [Fluency]
4. Explain why quadrupling the sample size halves a standard error. [Reasoning]
5. Simulate sample means from a skewed distribution for $n = 1, 2, 5, 30$. Quantify convergence rather than only describing the pictures. [Exploration]
6. Compare Monte Carlo error at $n = 100$, 10^4, and 10^6. Is the $n^{-1/2}$ rate visible? [Exploration]
7. Give an example where bootstrap resampling cannot reveal a sampling bias. [Reasoning]

Chapter 8

Learning from a sample

Estimation, confidence intervals, and hypothesis tests

Guiding question 8.1. How can a random sample support a calibrated statement about an unknown population?

An estimator $T = T(X_1, \dots, X_n)$ targets a parameter θ. Its bias, variance, and mean squared error are

$$\mathrm{Bias}_\theta(T) = \mathbb{E}_\theta[T] - \theta, \qquad \mathrm{MSE}_\theta(T) = \mathrm{Var}_\theta(T) + \mathrm{Bias}_\theta(T)^2.$$

Unbiasedness is attractive, but MSE captures the bias–variance tradeoff.

8.1 Method of moments and likelihood

The method of moments matches sample moments to model moments. Maximum likelihood chooses the parameter under which the observed data have greatest likelihood.

For Bernoulli observations $x_1, \dots, x_n$, the likelihood is

$$L(p) = p^{\sum x_i}(1-p)^{n-\sum x_i},$$

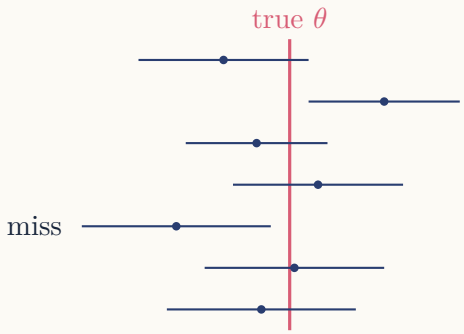
and the maximizer is $\hat{p} = \bar{x}$.

8.2 Confidence intervals

An approximate 95% interval for a mean with known or well-estimated standard error is

$$\text{estimate} \ \pm \ 1.96 \times \text{standard error}.$$

The confidence level refers to the long-run success rate of the procedure, not a posterior probability that a fixed parameter lies in one realized interval.



8.3 Tests and p-values

A hypothesis test specifies a null model H_0, a test statistic, and a rejection rule. The p-value is the probability, assuming H_0, of a result at least as incompatible with H_0 as the observed one.

It is not the probability that H_0 is true. A Type I error rejects a true null; a Type II error fails to reject a false null. Power is the probability of rejecting under a specified alternative.

Common trap

Statistical significance is not practical importance. With enough data, a tiny effect can yield a tiny p-value; with too little data, an important effect can be missed.

8.4 Model checking and design

Random sampling supports generalization; random assignment supports causal comparison. No inferential formula compensates for selection bias, poorly measured outcomes, hidden multiplicity, or a model contradicted by the data.

Exercises

1. Show that the sample mean is unbiased for μ. [Reasoning]
2. Find the Bernoulli maximum-likelihood estimator. [Fluency]
3. Compare two estimators when one has small bias but much smaller variance. [Reasoning]
4. Construct a 95% normal interval for a proportion from 600 observations with 330 successes. [Fluency]
5. Explain the repeated-sampling meaning of that interval without saying there is a 95% chance the fixed parameter is inside. [Reasoning]
6. A test reports $p = .03$. List three claims that do not follow. [Reasoning]
7. Simulate confidence-interval coverage and test power as functions of sample size. [Exploration]
8. Describe how optional stopping or multiple testing can inflate false discoveries. [Exploration]

Chapter 9

Probability for unknowns

An introduction to Bayesian inference

Chapter 10 will revisit this chapter's shared-parameter models from another direction: under infinite exchangeability, a conditional-i.i.d. representation is a theorem rather than a starting assumption.

Guiding question 9.1. What changes when an unknown parameter is itself represented by a probability distribution?

Bayesian inference combines a prior distribution $\pi(\theta)$ with a likelihood $p(y \mid \theta)$. Bayes' rule produces the posterior

$$\pi(\theta \mid y) = \frac{p(y \mid \theta)\pi(\theta)}{\int p(y \mid u)\pi(u)\,du} \propto p(y \mid \theta)\pi(\theta).$$

9.1 Beta–binomial updating

Suppose $p \sim \mathrm{Beta}(\alpha, \beta)$ and, conditional on p, $X \sim \mathrm{Bin}(n, p)$. After observing x successes,

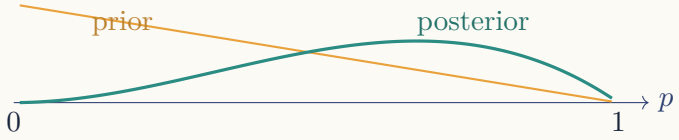
$$p \mid X = x \sim \mathrm{Beta}(\alpha + x, \beta + n - x).$$

The prior parameters behave like pseudocounts: $\alpha - 1$ prior successes and $\beta - 1$ prior failures when both exceed one.

The posterior mean is

$$\mathbb{E}[p \mid x] = \frac{\alpha + x}{\alpha + \beta + n},$$

a weighted compromise between the prior mean and the sample proportion.



9.2 Credible intervals and prediction

A 95% posterior credible interval contains 95% of the posterior probability. This is a conditional probability statement about the parameter under the model and prior. A posterior predictive

distribution averages future outcomes over posterior uncertainty:

$$p(y_{\text{new}} \mid y) = \int p(y_{\text{new}} \mid \theta)\pi(\theta \mid y)\, d\theta.$$

9.3 Prior sensitivity and checking

Priors make assumptions visible; they do not make assumptions disappear. Compare reasonable priors, examine posterior predictive simulations, and ask whether the fitted model reproduces important features of the data.

Big idea

Frequentist and Bayesian intervals answer different probability questions. The productive comparison is not "which formula looks similar?" but "what is random, what is conditioned upon, and what operating guarantee or posterior statement is being claimed?"

Exercises

1. Update a Beta$(2, 2)$ prior after 8 successes in 10 trials. [Fluency]
2. Compute the posterior mean and compare it with the sample proportion. [Fluency]
3. Derive beta–binomial conjugacy by collecting powers of p and $1 - p$. [Reasoning]
4. Under a Beta(α, β) posterior, find the predictive probability that the next Bernoulli trial succeeds. [Reasoning]
5. Compare Beta$(1, 1)$, Beta$(2, 2)$, and Beta$(20, 20)$ priors for the same small dataset. [Exploration]
6. Explain why a credible interval and confidence interval can have the same numerical endpoints but different interpretations. [Reasoning]
7. Design a posterior predictive check for a coin model that might have time-varying bias. [Exploration]

Capstone: one question, two inferential languages

Choose a binomial, count, or measurement dataset. State a generative model, inspect the data, estimate the target, and quantify uncertainty both frequentistically and Bayesianly. End with a model check and a paragraph explaining which conclusions are robust to the modeling choices.

Chapter 10

Exchangeability: when symmetry produces a parameter

A capstone on dependence, learning, and long-run frequency

Guiding question 10.1. If the order of observations carries no information, must the observations be independent?

No. But for an *infinite* sequence of zeros and ones, a remarkably precise substitute is available: the sequence behaves as though a success probability were chosen once, and independent trials were then run with that probability. This is de Finetti's representation theorem.

This chapter joins three earlier ideas: conditioning can remove dependence, Bayes' rule updates an unknown parameter, and repeated observations reveal a long-run frequency. Read the examples first. Section 10.9 is an optional proof, using a birthday-collision bound and one explicitly stated compactness fact from analysis.

10.1 Symmetry is weaker than independence

Definition 10.2. A finite random vector $(X_1, \dots, X_n)$ is *exchangeable* if permuting its coordinates does not change its joint distribution. An infinite sequence $X_1, X_2, \dots$ is exchangeable if every finite prefix is exchangeable.

For a zero–one vector, the definition says that two particular strings with the same number of ones have the same probability. Thus 1100, 1010, and 0011 must be equally likely. It does *not* say that 1100 and 1111 are equally likely, or that the coordinates are independent.

Independent and identically distributed (i.i.d.) variables are exchangeable: permuting a product of identical marginal laws changes nothing. Exchangeability also implies identical marginal distributions, because we can swap any two positions. Neither implication reverses in general.

Example 10.3 (One coin, chosen once)**.** Choose a coin with heads probability $1/4$ or $3/4$, each with probability $1/2$. Keep the chosen coin and toss it repeatedly, independently *given the choice*. Write $X_i = 1$ for heads.

For a specified length-n string containing s heads, its probability is

$$\frac{1}{2}\left(\frac{1}{4}\right)^{s}\left(\frac{3}{4}\right)^{n-s}+\frac{1}{2}\left(\frac{3}{4}\right)^{s}\left(\frac{1}{4}\right)^{n-s}.$$

Only the count s matters, so the sequence is exchangeable. Nevertheless,

$$\mathbb{P}(X_1=1)=\frac{1}{2},\qquad \mathbb{P}(X_1=X_2=1)=\frac{1}{2}\frac{1}{16}+\frac{1}{2}\frac{9}{16}=\frac{5}{16}\neq\frac{1}{4}.$$

Seeing a head makes the high-bias coin more plausible, and therefore makes another head more likely. The tosses are dependent when the coin's identity is unknown.

Checkpoint

If we independently choose a fresh coin before *each* toss, the results are instead i.i.d. fair bits. Sharing one random choice across all trials is the source of dependence in Example 10.3.

10.2 The Bernoulli representation theorem

Theorem 10.4 (de Finetti, Bernoulli case)**.** *An infinite zero–one sequence is exchangeable if and only if there is a probability measure μ on $[0,1]$ such that, for every $n\geq 1$ and every $(x_1,\dots,x_n)\in\{0,1\}^n$ with $s=\sum_i x_i$,*

$$\mathbb{P}(X_1=x_1,\dots,X_n=x_n)=\int_{[0,1]}p^{s}(1-p)^{n-s}\,\mu(dp). \tag{10.1}$$

The measure μ is unique. Equivalently, the sequence has the law of conditionally i.i.d. Bernoulli(Θ) trials, where Θ is chosen once with distribution μ.

An exponent zero contributes a factor of one, including at $p=0$ or 1. The integral averages over a probability *measure*: μ may have a density, point masses, or both. It need not be a beta distribution. We call μ the *mixing law* or *de Finetti measure*.

Two levels of randomness

First choose $\Theta\sim\mu$. Conditional on $\Theta=p$, generate $X_1,X_2,\dots$ independently with success probability p. Conditional independence concerns the second level; exchangeability describes the joint law after averaging over the first.

The easy direction is worth proving now: every integrand in (10.1) depends on the string only through s, so every mixture is exchangeable. The difficult direction says that *every infinite exchangeable Bernoulli law* admits such a representation.

Let $S_n=X_1+\cdots+X_n$. There are $\binom{n}{s}$ strings with s ones, so

$$\mathbb{P}(S_n=s)=\binom{n}{s}\int_{[0,1]}p^{s}(1-p)^{n-s}\,\mu(dp). \tag{10.2}$$

The binomial coefficient belongs to the *count* probability, not the probability of one specified ordered string. This distinction is exactly the permutations-and-combinations distinction from Chapter 1.

Ordinary i.i.d. Bernoulli(p_0) trials correspond to $\mu=\delta_{p_0}$, a point mass at p_0. At the other extreme, choose a fair bit Z and put $X_i=Z$ for every i. The sequence is exchangeable, all trials agree, and $\mu=\frac{1}{2}\delta_0+\frac{1}{2}\delta_1$.

What the theorem does not choose

Exchangeability specifies a class of models, not a particular prior. Uniqueness means that the *entire infinite joint law* determines μ; it does not mean that finitely many observations determine it. Nor does the representation prove that nature literally selected a hidden coin. Different mechanisms can have exactly the same observable law.

10.3 Learning about the shared coin

Return to Example 10.3. After a specified string with s heads in n tosses, Bayes' rule gives

$$q_{n,s} := \mathbb{P}\left(\Theta = \frac{3}{4} \,\middle|\, X_1 = x_1, \dots, X_n = x_n\right) = \frac{3^s}{3^s + 3^{n-s}}.$$

The probability of heads on the next toss is therefore

$$\mathbb{P}(X_{n+1} = 1 \mid X_1 = x_1, \dots, X_n = x_n) = \frac{1}{4}(1 - q_{n,s}) + \frac{3}{4} q_{n,s}.$$

After two heads, $q_{2,2} = 9/10$ and the predictive probability is $7/10$. After a head and a tail, $q_{2,1} = 1/2$ and it is $1/2$. The order HT versus TH does not affect either answer.

For an arbitrary mixing law, the same calculation gives

$$\mathbb{P}(X_{n+1} = 1 \mid X_1 = x_1, \dots, X_n = x_n) = \frac{\int p^{s+1}(1-p)^{n-s}\,\mu(dp)}{\int p^{s}(1-p)^{n-s}\,\mu(dp)}, \tag{10.3}$$

provided the observed string has positive probability. This is the posterior mean of Θ. Conditioning on the count $S_n = s$ gives the same result, since the factor $\binom{n}{s}$ cancels.

There is no conflict between exchangeability and learning. Exchangeability says that the *order* of the observed bits carries no additional information beyond their count. It does not say that the count carries no information about future bits.

10.4 Pólya's urn: reinforcement becomes a mixture

Start with one red and one blue ball. Draw a ball uniformly, replace it, and add one extra ball of the same color. Repeat. Let $X_i = 1$ mean that draw i is red. After s red draws among n draws, the next draw is red with probability

$$\frac{s+1}{n+2}. \tag{10.4}$$

This mechanism seems to remember its past: drawing red makes future reds more likely. Yet its sequence is exchangeable.

Example 10.5 (Calculate an ordered path)**.** The path $RRRB$ has probability

$$\frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{1}{5} = \frac{1}{20}.$$

The path $RBRR$ has probability $\frac{1}{2} \cdot \frac{1}{3} \cdot \frac{2}{4} \cdot \frac{3}{5} = 1/20$ as well. These calculations are not the probability of getting three reds in four draws: there are four possible positions for the blue draw, giving $4/20 = 1/5$ for that event.

For any specified path with s red and $n-s$ blue draws, the red numerators, in their order of appearance, are $1, 2, \ldots, s$; the blue numerators are $1, 2, \ldots, n-s$. The denominators are $2, 3, \ldots, n+1$. Consequently

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \frac{s!(n-s)!}{(n+1)!}. \tag{10.5}$$

The order changes where the factors appear, but not their product.

Repeated integration by parts gives the elementary beta integral

$$\int_0^1 p^s (1-p)^{n-s}\, dp = \frac{s!(n-s)!}{(n+1)!}.$$

Comparing this with (10.1) identifies the mixing law: Θ is uniform on $[0,1]$. Thus two different descriptions generate the same law of observed colors:

1. reinforce the color just drawn, using the urn rule;

2. choose one uniform random bias Θ, then generate conditionally independent Bernoulli(Θ) draws.

This is equality of joint distributions, not a claim that the physical urn contains a concealed coin.

An especially striking consequence is

$$\mathbb{P}(S_n = s) = \binom{n}{s} \frac{s!(n-s)!}{(n+1)!} = \frac{1}{n+1}, \qquad s = 0, \ldots, n.$$

The *count* is uniform, even though the individual strings are not. For four draws, all five counts $0, 1, 2, 3, 4$ have probability $1/5$. Compare independent fair tosses, whose count probabilities are $1, 4, 6, 4, 1$, divided by 16.

Other starting compositions

With a red and b blue balls initially, where a, b are positive integers, define the rising factorial $a^{\overline{r}} = a(a+1)\cdots(a+r-1)$ and $a^{\overline{0}} = 1$. It is different from the falling factorial used to count ordered selections without replacement. The path probability becomes

$$\frac{a^{\overline{s}} b^{\overline{n-s}}}{(a+b)^{\overline{n}}} = \int_0^1 p^s (1-p)^{n-s} \frac{p^{a-1}(1-p)^{b-1}}{B(a,b)}\, dp,$$

where $B(a,b) = \int_0^1 p^{a-1}(1-p)^{b-1}\, dp$. The mixing law is Beta$(a,b)$.

The Bayesian model also makes sense for arbitrary real $a, b > 0$, whether or not we picture whole balls. After s successes in n trials, the posterior is Beta$(a+s, b+n-s)$ and the predictive success probability is $(a+s)/(a+b+n)$. For the one-red, one-blue urn, this reproduces (10.4). Chapter 9's beta update now has a concrete reinforcement interpretation.

10.5 A law of large numbers with a random limit

For fixed p, the strong law for independent Bernoulli(p) trials says $S_n/n \to p$ with probability one. This means that the set of infinite sample paths where convergence fails has probability zero; it is stronger than convergence in probability in Chapter 7. A short proof of this special strong law appears in Section 10.9.

Apply the strong law conditional on the shared bias. It gives

$$\frac{S_n}{n} \longrightarrow \Theta \qquad \text{with probability one.} \tag{10.6}$$

In particular, the mixing law is the distribution of the limiting empirical frequency. The parameter is not merely an algebraic trick: it can be recovered from the infinite sequence itself.

Example 10.6 (Same marginal, different limits)**.** In each of these models $\mathbb{P}(X_i = 1) = 1/2$:

Model	Limit of S_n/n
Independent fair tosses	1/2, with probability one
One coin of bias 1/4 or 3/4	1/4 or 3/4, each with probability 1/2
One-red, one-blue Pólya urn	A uniform random value in $[0, 1]$
One fair bit repeated forever	0 or 1, each with probability 1/2

Identical one-trial probabilities tell us very little about the behavior of a long sequence.

We can see the persistent variation without invoking a limit theorem. Conditional independence and the tower property give, for $i \neq j$,

$$\mathbb{E}X_i = \mathbb{E}\Theta, \qquad \mathbb{E}[X_iX_j] = \mathbb{E}[\Theta^2], \qquad \boxed{\mathrm{Cov}(X_i, X_j) = \mathrm{Var}(\Theta) \geq 0.}$$

Also, conditional on Θ, the sample mean has variance $\Theta(1-\Theta)/n$ and mean Θ. Taking expectations in the decomposition into within-group and between-group variation gives

$$\boxed{\mathrm{Var}(S_n/n) = \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \mathrm{Var}(\Theta).} \tag{10.7}$$

For the two-coin model this is $3/(16n) + 1/16$; for the one-red, one-blue urn it is $1/(6n) + 1/12$.

Increasing the number of observations reduces the sampling noise *within a run*. It does not remove the variation of the shared bias *between runs*. In the urn example, for instance, $\mathbb{P}(\lim S_n/n > 3/4) = 1/4$, despite the perfectly symmetric starting urn.

Explore: one long run versus many runs

Simulate 100 independent copies of the one-red, one-blue urn, each for 5,000 draws. Plot a few running red frequencies; then make a histogram of the 100 final frequencies. Compare with 100 runs of independent fair tosses and with 100 runs of the two-coin model. Predict all three histograms before simulating. A single long trajectory estimates its own shared bias, not the distribution of biases across runs.

10.6 Why infinity is essential

Example 10.7 (A symmetric pair that cannot continue)**.** Let (X_1, X_2) be $(1, 0)$ or $(0, 1)$, each with probability $1/2$. This pair is exchangeable, but

$$\mathrm{Cov}(X_1, X_2) = 0 - \frac{1}{2}\frac{1}{2} = -\frac{1}{4}.$$

It cannot be a mixture of conditionally i.i.d. Bernoulli trials, since every such mixture has nonnegative covariance. In fact, it cannot even be the first two coordinates of an exchangeable triple: all three pairs would have to disagree with probability one, which three bits cannot do.

More generally, sample without replacement from N balls, M red and $N - M$ blue. The N color indicators are exchangeable, but for $i \neq j$ and $p = M/N$,

$$\mathbb{P}(X_i = X_j = 1) = \frac{M(M-1)}{N(N-1)}, \qquad \mathrm{Cov}(X_i, X_j) = -\frac{p(1-p)}{N-1} \quad (N > 1).$$

For $0 < M < N$, exhausting red balls creates negative dependence. Finite symmetry is real, but it need not be the symmetry of an infinite sequence.

The finite bridge

Conditional on its total number of ones, every finite exchangeable zero–one vector is a uniform arrangement of those ones and zeros. It is therefore a mixture of sampling-without-replacement models. For a short sample from a large population, replacement makes little difference. This observation is the bridge to de Finetti's theorem.

10.7 Beyond coins, and the limits of symmetry

The corresponding theorem for an infinite exchangeable sequence of *real-valued* random variables replaces the random number Θ by a random probability distribution Q on $\mathbb{R}$. Conditional on Q, the variables are i.i.d. with law Q. For Borel sets $A_1, \ldots, A_n$,

$$\mathbb{P}(X_1 \in A_1, \ldots, X_n \in A_n) = \mathbb{E}\left[\prod_{i=1}^{n} Q(A_i)\right].$$

This general result is stated for perspective, not proved here. The real-valued setting matters: unrestricted measurable spaces require additional care.

For example, let M be a random group mean and let $X_i = M + \varepsilon_i$, where the ε_i are i.i.d. $\mathcal{N}(0, 1)$ and independent of M. Conditional on M, the observations are i.i.d. $\mathcal{N}(M, 1)$; unconditionally, they are exchangeable and, when $\mathrm{Var}(M) > 0$ is finite, have covariance $\mathrm{Var}(M) > 0$. Here $Q = \mathcal{N}(M, 1)$ is a random distribution.

Exchangeability is a modeling assumption to defend. Time trends, different treatment groups, and changing sampling protocols can make permutations inappropriate. Declaring observations exchangeable does not erase such structure. Its value is to make a precise connection: *a symmetry of the observable law can justify a conditional i.i.d. representation, without assuming unconditional independence.*

10.8 Exercises

Exchangeability practice

1. In the two-coin model, observe three heads and one tail. Find the posterior probability that the bias is $3/4$ and the predictive probability of another head. Would $HTHH$ give a different answer? [Fluency]
2. Prove that independently selecting a fresh bias from $\{1/4, 3/4\}$ before every toss produces i.i.d. fair bits. Identify which part of the shared-coin model has changed. [Reasoning]
3. For the one-red, one-blue urn, find the probability of the specified path $RRBRR$, and then the probability of exactly four reds in five draws. Given this path, find the probability of a red next. [Fluency]
4. An urn starts with two red and three blue balls and uses Pólya reinforcement. After four red and two blue draws, identify the mixing law before observing data, the posterior law, and the next-red probability. [Fluency]
5. Derive (10.7). Under uniform mixing, compute $\mathrm{Var}(S_{10}/10)$ and $\mathrm{Var}(S_{100}/100)$. Explain why neither should be approximated by $1/(4n)$. [Reasoning]
6. Suppose an infinite exchangeable Bernoulli sequence is pairwise independent. Prove that it is i.i.d. What hypothesis prevents this from contradicting Chapter 3's pairwise-independence warning? [Reasoning]
7. Let an exchangeable vector of length $N \geq 2$ have common finite variance v and common pairwise covariance c. Prove $c \geq -v/(N-1)$. Use this to rule out an exchangeable extension of the always-disagreeing pair to any length $N > 2$. [Reasoning]
8. Run the experiment in the Explore box. For each model, state whether final frequencies from many independent runs should cluster near one point, two points, or across an interval. Separate a simulation observation from a theorem that explains it. [Exploration]
9. Use the coupling in Section 10.9 to bound the error for *any event* concerning the first five draws from an exchangeable zero–one population of size 1,000. Explain why the same bound need not be informative for 100 draws. [Optional proof]

Sources and a next step

The representation theorem is due to Bruno de Finetti. For another elementary Bernoulli proof, using moments, see Werner Kirsch, *An elementary proof of de Finetti's Theorem* (2018). Finite exchangeability and sampling approximations are treated systematically in Persi Diaconis and David Freedman, *Finite Exchangeable Sequences* (1980).

10.9 Optional proof: finite urns and birthday collisions

Here is a proof of the Bernoulli representation, its uniqueness, and the random-frequency interpretation. The finite approximation is elementary. We isolate the one compactness input rather than conceal it in a limit.

Step 1: condition on the finite total

Fix $n \le N$ and a particular length-n string with s ones. Conditional on $S_N = k$, exchangeability makes all $\binom{N}{k}$ length-N strings with k ones equally likely, whenever $\mathbb{P}(S_N = k) > 0$. Hence

$$\mathbb{P}(X_1 = x_1, \dots, X_n = x_n \mid S_N = k) = \frac{\binom{N-n}{k-s}}{\binom{N}{k}}.$$

A binomial coefficient is zero when its lower argument is outside its allowed range. This is precisely the probability of that ordered color string when sampling without replacement from k ones and $N - k$ zeros.

Step 2: allow replacement and bound the error

Draw n independent uniform labels $J_1, \dots, J_n$ from $\{1, \dots, N\}$. If no label repeats, use these same labels as the without-replacement sample. If a label repeats, generate a new uniform ordered sample of distinct labels for the without-replacement sample. This construction has the correct marginal sampling laws and makes their color strings identical whenever the original labels are distinct.

For each pair $i < j$, $\mathbb{P}(J_i = J_j) = 1/N$. The union bound from the birthday problem gives

$$\mathbb{P}(\text{a repeated label}) \le \min\left\{1, \frac{\binom{n}{2}}{N}\right\}. \tag{10.8}$$

Two coupled random strings that agree except on an event of probability ε assign probabilities differing by at most ε to *any* event: their event indicators differ only when the strings differ. With replacement, the colors are independent Bernoulli(k/N). Thus, writing μ_N for the distribution of S_N/N, and averaging over k,

$$\left|\mathbb{P}(X_1 = x_1, \dots, X_n = x_n) - \int p^s (1-p)^{n-s}\, \mu_N(dp)\right| \le \frac{\binom{n}{2}}{N}. \tag{10.9}$$

The left-hand probability is fixed as N grows: all prefixes belong to the *same* infinite sequence.

Step 3: take a subsequence of probability laws

Lemma 10.8 (Compactness on a closed bounded interval)**.** *Every sequence of probability measures on $[0,1]$ has a subsequence ν_j and a probability measure ν on $[0,1]$ such that $\int f\, d\nu_j \to \int f\, d\nu$ for every continuous function $f : [0,1] \to \mathbb{R}$. Also, two probability measures with equal integrals against all such functions are equal.*

This is the analysis input, also called weak sequential compactness. One route to its proof selects convergent subsequences of distribution functions at each rational point, then uses monotonicity to

obtain a limiting distribution function at its continuity points. Compact support prevents probability from escaping to infinity. We use the lemma here without developing that analysis.

Apply the lemma to μ_N. Along one subsequence, μ_{N_j} converges weakly to some μ. For fixed n and s, the function $p^s(1-p)^{n-s}$ is continuous, so (10.9) yields (10.1). The same subsequence works for every continuous test function, hence for every finite string. This proves existence.

Step 4: show that the mixing law is unique

Under *any* candidate mixture, conditional on $\Theta = p$, $S_n \sim \mathrm{Bin}(n, p)$. Chebyshev gives the uniform bound

$$\mathbb{P}(|S_n/n - p| > \delta \mid \Theta = p) \le \frac{1}{4n\delta^2} \qquad (\delta > 0).$$

For a continuous f on $[0,1]$, let $K = \sup|f|$ and $\omega_f(\delta) = \sup_{|x-y|\le\delta}|f(x) - f(y)|$. Split according to whether $|S_n/n - \Theta| \le \delta$ to get

$$|\mathbb{E}f(S_n/n) - \mathbb{E}f(\Theta)| \le \omega_f(\delta) + \frac{2K}{4n\delta^2}.$$

First let $n \to \infty$, then $\delta \downarrow 0$; uniform continuity makes the right-hand side vanish. Thus the distribution of S_n/n converges weakly to the mixing law. The observable finite-dimensional laws already determine the left-hand side, so two candidate mixing laws have equal continuous-function integrals and are equal by Lemma 10.8.

Step 5: justify the almost-sure random limit

We first prove the strong law for fixed p without assuming it. For i.i.d. Bernoulli(p) trials, Chebyshev along square indices gives

$$\mathbb{P}\left(\left|\frac{S_{m^2}}{m^2} - p\right| > \varepsilon\right) \le \frac{1}{4m^2\varepsilon^2}.$$

The probability that at least one such deviation occurs for some $m \ge M$ is at most $\sum_{m\ge M}(4m^2\varepsilon^2)^{-1}$, which tends to zero. Therefore only finitely many deviations occur with probability one. Intersecting these probability-one conclusions for $\varepsilon = 1, 1/2, 1/3, \ldots$ proves $S_{m^2}/m^2 \to p$ almost surely.

Since the partial sums are nondecreasing, for $m^2 \le n \le (m+1)^2$,

$$\frac{S_{m^2}}{(m+1)^2} \le \frac{S_n}{n} \le \frac{S_{(m+1)^2}}{m^2}.$$

Both bounds tend to p, proving $S_n/n \to p$ almost surely.

Now construct the mixture by choosing $\Theta \sim \mu$ and then the conditionally independent trials. The fixed-p result has conditional probability one for every p, so averaging gives $S_n/n \to \Theta$ with probability one. The constructed sequence has the same finite-string probabilities as the original sequence. These determine the probability law on infinite zero–one sequences (including convergence events, which use countable unions and intersections). Thus the original sequence also has an almost-sure empirical limit, distributed as μ.

Taking that limit as the original sequence's Θ transfers the conditional-i.i.d. representation as well: the pair consisting of the sequence and its empirical limit has the same law in the two descriptions. This completes the promised interpretation of the theorem.

Appendix A

Selected hints and solutions

Chapter 1

For the at-least-one-ace problem, count the complement: $\binom{52}{5} - \binom{48}{5}$. For divisibility by 2, 3, or 5, inclusion–exclusion gives

$$500 + 333 + 200 - 166 - 100 - 66 + 33 = 734.$$

The four-person selection counts from nine students are $\binom{9}{4} = 126$, $(9)_4 = 3024$, and $4\binom{9}{4} = 504$. The repeated-letter count for `LEVEL` is $5!/(2!2!) = 30$. The coefficient of x^2 in $(1+3x)^5$ is $\binom{5}{2}3^2 = 90$. The length-five strings using all three symbols number $3^5 - 3 \cdot 2^5 + 3 = 150$.

Chapter 2

For the maximum of two dice to equal 4, both dice must be at most 4 but not both at most 3, giving $(4^2 - 3^2)/36 = 7/36$. For two uniform points X, Y, the band $|X - Y| < .1$ has area $1 - 2(.9)^2/2 = .19$. For the weighted four-outcome space, the probability of an even outcome is $(2+4)/10 = 3/5$, not $2/4$. On a 30-day calendar, the birthday collision probability is about .41356 for six people and .53084 for seven; thus the first size above $1/2$ is seven. For four people on five days, there are $5 \cdot 4 \cdot 3 \cdot 2 = 120$ distinct-birthday lists among $5^4 = 625$ lists, so the collision probability is $1 - 120/625 = 101/125$. The quantity $\binom{n}{2}/d$ sums probabilities of overlapping events. It is an upper bound, not necessarily a probability.

Chapter 3

For the coin problem,

$$\mathbb{P}(D \mid HHH) = \frac{1 \cdot .2}{1 \cdot .2 + (1/2)^3 \cdot .8} = \frac{2}{3}.$$

A pairwise-not-mutual example: flip two fair coins; let A be first-heads, B second-heads, and C equal-results.

Chapter 4

For fixed points, write $N = \sum_{i=1}^{n} I_i$. Since $\mathbb{P}(I_i = 1) = 1/n$, linearity gives $\mathbb{E}N = 1$. The variance identity follows by expanding $(X - \mu)^2$.

Chapter 5

For an exponential variable, $\mathbb{P}(X > m) = e^{-\lambda m} = 1/2$, hence $m = (\log 2)/\lambda$. For $X \sim \mathcal{N}(10, 9)$,

$$\mathbb{P}(8 < X < 13) = \Phi(1) - \Phi(-2/3).$$

Chapter 6

For X uniform on $\{-1, 0, 1\}$ and $Y = X^2$, symmetry gives $\mathbb{E}X = \mathbb{E}[XY] = 0$, hence covariance zero, but knowing X determines Y. For the tower property, take the weighted average of group means; each observation receives exactly its population weight.

Chapter 7

If $\sigma = 12, n = 64$, then $\mathrm{SE}(\bar{X}) = 12/8 = 1.5$. The weak law follows from $\mathrm{Var}(\bar{X}_n) = \sigma^2/n$ and Chebyshev.

Chapter 8

For 330 successes in 600 trials, $\hat{p} = .55$ and the plug-in standard error is

$$\sqrt{.55(.45)/600} \approx .0203.$$

The approximate 95% interval is $.55 \pm 1.96(.0203)$, or about $(.510, .590)$.

Chapter 9

Beta$(2, 2)$ updated with 8 successes and 2 failures becomes Beta$(10, 4)$, with posterior mean $10/14 \approx .714$. The posterior predictive success probability equals the posterior mean $\alpha'/(\alpha' + \beta')$.

Chapter 10

The numbering here matches Section 10.8.

1. The posterior probability of bias $3/4$ is $3^3/(3^3+3^1) = 9/10$. The predictive chance of heads is $(1/4)(1/10) + (3/4)(9/10) = 7/10$. Reordering the observed string does not change either calculation.

2. Let the newly chosen biases be independent variables $\Theta_1, \ldots, \Theta_n$. For any specified bits,

$$\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) = \mathbb{E}\prod_{i=1}^{n} \Theta_i^{x_i}(1-\Theta_i)^{1-x_i} = \prod_{i=1}^{n} \frac{1}{2} = 2^{-n}.$$

Here the tosses are generated independently given all the biases. The expectation factors because the biases are independent; it does not factor this way when a single shared Θ is reused.

3. The ordered path has probability $4!1!/6! = 1/30$. There are $\binom{5}{4} = 5$ such paths, so exactly four reds has probability $1/6$. After the path, the urn contains five red and two blue balls; the next-red probability is $5/7$.

4. Initially $\Theta \sim \text{Beta}(2,3)$. The posterior is $\text{Beta}(6,5)$ and the predictive probability is $6/11$. The updated urn also contains six red and five blue balls.

5. Write $Y = S_n/n$ and $m = \mathbb{E}\Theta$. Expand $Y - m = (Y - \Theta) + (\Theta - m)$. The cross term has expectation zero because $\mathbb{E}[Y - \Theta \mid \Theta] = 0$. Therefore

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y \mid \Theta)] + \text{Var}(\Theta) = \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \text{Var}(\Theta).$$

Under uniform mixing, $\mathbb{E}\Theta = 1/2$, $\mathbb{E}\Theta^2 = 1/3$, so the two terms are $1/(6n)$ and $1/12$. This gives $\text{Var}(S_{10}/10) = 1/10$ and $\text{Var}(S_{100}/100) = 17/200$. The expression $1/(4n)$ wrongly treats the trials as independent Bernoulli$(1/2)$ variables; it misses the between-run variance.

6. Pairwise independence implies $\text{Cov}(X_1, X_2) = 0$, hence $\text{Var}(\Theta) = 0$ by the representation theorem. Thus $\Theta = p_0$ almost surely for some constant p_0, and (10.1) is the i.i.d. Bernoulli(p_0) product law. Infinite exchangeability is the additional hypothesis. An arbitrary finite collection of pairwise independent events need not have it.

7. Nonnegativity of variance gives

$$0 \le \text{Var}\left(\sum_{i=1}^{N} X_i\right) = Nv + N(N-1)c.$$

Divide by $N(N-1)$ to obtain the bound. The always-disagreeing pair has $v = 1/4, c = -1/4$, so an exchangeable extension would require $-1/4 \ge -1/[4(N-1)]$, possible only for $N \le 2$.

8. Fair-toss frequencies cluster near $1/2$; shared-coin frequencies cluster near $1/4$ and $3/4$; urn frequencies spread across $[0,1]$. Finite simulations fluctuate, and 100 runs will not give a perfectly flat histogram. The theorem identifies the limiting laws as $\delta_{1/2}$, $\frac{1}{2}\delta_{1/4} + \frac{1}{2}\delta_{3/4}$, and the uniform law, respectively.

9. Conditional on the total number of ones, compare sampling without replacement with independent sampling at that population's success proportion; then average over the total. For any event, the absolute probability error is at most $\binom{5}{2}/1000 = .01$. With 100 draws the union bound is $\binom{100}{2}/1000 = 4.95$; capped at one, it is only the trivial bound. This does not say the approximation error actually equals one. The collision argument is useful when n^2/N is small.

Appendix B

Reference sheets

Counting and overlap corrections

For $0 \le k \le n$, ordered selections without replacement number $(n)_k = n!/(n-k)!$; unordered selections number $\binom{n}{k} = n!/[k!(n-k)!]$. Ordered k-term lists with replacement number n^k. Here $0! = 1$.

$$\begin{aligned}
\binom{n}{k} &= \binom{n}{n-k}, \\
\binom{n}{k} &= \binom{n-1}{k-1} + \binom{n-1}{k}, \\
(x+y)^n &= \sum_{k=0}^{n} \binom{n}{k} x^{n-k} y^k, \\
|A \cup B \cup C| &= |A| + |B| + |C| - |A \cap B| - |A \cap C| \\
&\quad - |B \cap C| + |A \cap B \cap C|.
\end{aligned}$$

The same alternating formula holds for event probabilities, without an independence assumption. Pairwise intersections include the triple region.

Finite probability spaces and birthdays

For finite Ω, assign weights $w_\omega \ge 0$ summing to one; then $\mathbb{P}(A) = \sum_{\omega \in A} w_\omega$. Only in the uniform case does this simplify to $|A|/|\Omega|$.

For n independent uniform birthdays on d days,

$$\mathbb{P}(\text{some birthday match}) = \begin{cases} 1 - \displaystyle\prod_{j=0}^{n-1} (1 - j/d), & 0 \le n \le d, \\ 1, & n > d. \end{cases}$$

For a designated person among $n \ge 1$, the probability of a match with someone else is $1-(1-1/d)^{n-1}$, taking it to be zero when $n = 1$. For 365 days, the any-pair probability first exceeds $1/2$ at 23 people.

Core probability identities

$$\begin{aligned}
\mathbb{P}(A^c) &= 1 - \mathbb{P}(A), \\
\mathbb{P}(A \cup B) &= \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B), \\
\mathbb{P}(A \mid B) &= \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}, \\
\mathbb{P}(A) &= \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \\
\mathbb{P}(B_j \mid A) &= \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)}.
\end{aligned}$$

Moments and dependence

$$\begin{aligned}
\mathbb{E}[aX + bY] &= a\mathbb{E}X + b\mathbb{E}Y, \\
\operatorname{Var}(X) &= \mathbb{E}[X^2] - (\mathbb{E}X)^2, \\
\operatorname{Cov}(X, Y) &= \mathbb{E}[XY] - \mathbb{E}X\mathbb{E}Y, \\
\operatorname{Var}(X + Y) &= \operatorname{Var} X + \operatorname{Var} Y + 2\operatorname{Cov}(X, Y), \\
\rho_{X,Y} &= \operatorname{Cov}(X, Y)/(\sigma_X \sigma_Y).
\end{aligned}$$

Common distributions

Distribution	Support	Mean	Variance
Bernoulli(p)	$\{0, 1\}$	p	$p(1-p)$
Binomial(n, p)	$0, \ldots, n$	np	$np(1-p)$
Geometric(p)	$1, 2, \ldots$	$1/p$	$(1-p)/p^2$
Poisson(λ)	$0, 1, \ldots$	λ	λ
Uniform(a, b)	$[a, b]$	$(a+b)/2$	$(b-a)^2/12$
Exponential(λ)	$[0, \infty)$	$1/\lambda$	$1/\lambda^2$
Normal(μ, σ^2)	$\mathbb{R}$	μ	σ^2
Beta(α, β)	$[0, 1]$	$\alpha/(\alpha+\beta)$	$\alpha\beta/[(\alpha+\beta)^2(\alpha+\beta+1)]$

Inference checklist

1. What is the target population and parameter?

2. How were units sampled or assigned?

3. What model connects observations to the parameter?

4. What is the estimator and its sampling uncertainty?

5. Which assumptions are needed for the interval or test?

6. Is the effect practically meaningful?

7. What diagnostic or predictive check could reveal model failure?

Exchangeability and a shared random bias

For an *infinite* exchangeable zero–one sequence, de Finetti's theorem gives a unique mixing law μ on $[0,1]$. Choose $\Theta \sim \mu$ once; conditional on Θ, the trials are i.i.d. Bernoulli(Θ). For a specified string with s ones,

$$\begin{aligned}
\mathbb{P}(X_1 = x_1, \ldots, X_n = x_n) &= \int p^s (1-p)^{n-s}\, \mu(dp), \\
\mathbb{P}(S_n = s) &= \binom{n}{s} \int p^s (1-p)^{n-s}\, \mu(dp), \\
\mathrm{Cov}(X_i, X_j) &= \mathrm{Var}(\Theta) \quad (i \neq j), \\
\mathrm{Var}(S_n/n) &= \frac{\mathbb{E}[\Theta(1-\Theta)]}{n} + \mathrm{Var}(\Theta), \\
S_n/n &\longrightarrow \Theta \quad \text{with probability one.}
\end{aligned}$$

The predictive success probability is the posterior mean of Θ. With a Beta(a,b) mixing law and s successes in n trials,

$$\Theta \mid \text{data} \sim \mathrm{Beta}(a+s, b+n-s), \qquad \mathbb{P}(X_{n+1} = 1 \mid \text{data}) = \frac{a+s}{a+b+n}.$$

Pólya's urn, with one-ball reinforcement and a, b initial balls of the two colors, realizes this model for positive integer a, b. For $a = b = 1$, the mixing law is uniform and S_n is uniform on $\{0, \ldots, n\}$.

Common trap

Finite exchangeability alone does not imply a mixture of i.i.d. trials. Sampling without replacement can have negative covariance. Exchangeability also does not choose a particular prior or identify a unique physical mechanism.

For a prefix of length $n \leq N$ from an exchangeable zero–one vector of length N, the finite-mixture approximation using bias S_N/N has absolute probability error at most $\min\{1, \binom{n}{2}/N\}$ for any event on that prefix. See Chapter 10 for the coupling proof.

Further reading

- D. Freedman, R. Pisani, and R. Purves, *Statistics*.
- J. Blitzstein and J. Hwang, *Introduction to Probability*.
- L. Wasserman, *All of Statistics* (for a concise next course).
- A. Gelman et al., *Bayesian Data Analysis* (for a more advanced Bayesian treatment).
- W. Kirsch, *An elementary proof of de Finetti's Theorem*, arXiv:1809.00882 (2018).
- P. Diaconis and D. Freedman, *Finite Exchangeable Sequences*, Annals of Probability **8**(4) (1980), 745–764.

Course packet designed as an adaptable teaching draft. Classical results and examples are attributed where appropriate; exposition, exercises, and diagrams are prepared for this packet.